Plan

- Directions of Fea(LPP)
- Extreme directions of Fea(LPP)
- Representation Theorem for Fea(LPP)
- Necessary and sufficient conditions for the existence of optimal solutions
- Optimal solutions in atleast one corner point

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- Directions $\mathbf{d}_1, \mathbf{d}_2$ of S are said to be distinct if $\mathbf{d}_1 \neq \gamma \mathbf{d}_2$ for any $\gamma > 0$
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- Whereas $\mathbf{d}_1 = [1, 1]^T$, $\mathbf{d}_2 = [1, 0]^T$ give two distinct directions

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- The number of distinct extreme directions of S is finite (why?)

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- **Theorem:**
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- The theorem works for $Fea(LPP)$ because of the non negativity constraints, and $Fea(LPP)$ has n LI , defining hyperplanes

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- **Definition:** Given S, a nonempty subset of $\mathbb{R}^n$, and $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k \in S$
  $\sum_{i=1}^{k} \lambda_i \mathbf{x}_i$, is called a convex combination of $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ where $0 \leq \lambda_i \leq 1$ for all $i = 1, 2, \ldots, k$, and $\sum_{i=1}^{k} \lambda_i = 1$

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- **Result:** Given $\phi \neq S \subset \mathbb{R}^n$, S is a convex set $\Leftrightarrow$ for all $k \in \mathbb{N}$,
  the convex combination of any k elements of S is in S.

- **Theorem: (Representation Theorem)**
  If $S = \{\mathbf{x} \in \mathbb{R}^n : A_{m\times n}\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ is nonempty and if $\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_k$ are the extreme points of S and $\mathbf{d}_1, \mathbf{d}_2, .., \mathbf{d}_r$ are the distinct extreme directions of S (the set of directions is empty if S is bounded) then $\mathbf{x} \in S \quad \Leftrightarrow$

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- Since D', the set of distinct directions of S (if it is nonempty) is bounded ($\mathbf{d} \geq \mathbf{0}$ and $\sum_{i=1}^{n} d_i = 1$)
  so any $\mathbf{d} \in D'$ is a convex combination of the extreme points of D'

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- **Exercise:** Give an example of a **nonlinear** function $f : S \to \mathbb{R}$, where $S \subset \mathbb{R}$ is a closed and bounded polyhedral subset of $\mathbb{R}$, (what are these sets?) such that f has a minimum value in S but the minimum value is not attained at any extreme point of S

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